



Multi-physics modeling of keyhole-induced bubble formation dynamics: Effects of melt pool characteristics in laser metal processing

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Highlights

- High surface tension delays the transition time, thereby suppressing bubble-forming.
- Competition of surface tension and recoil pressure is quantitatively characterized.
- Smaller beam promotes bubble formation with a high aspect ratio of the melt pool.
- Bubble formation is related to the depth ratio during the keyhole fluctuation period.

Abstract

Porosity is a fundamental barrier to defect-free laser processing, originating from bubble formation. In light of the current lack of sufficient research in bubble formation, this study employs a 3D multi-phase thermo-fluidic model to investigate keyhole closure and bubble formation mechanisms. Simulations reveal that shutting off the laser during keyhole mode readily induces bubble formation, whereas conduction mode produces no bubbles owing to greater keyhole stability. By increasing the surface tension, the transition to keyhole mode can be delayed, thereby postponing or reducing the risk of bubble formation. Accordingly, the study quantified the transition time distribution within the parameter space of surface tension and recoil pressure, confirming that increasing surface tension or reducing recoil pressure delays the mode transition, thereby suppressing bubble formation. Furthermore, by independently modulating laser process parameters, this study systematically investigates the influence of melt pool geometric parameters on bubble formation mechanisms. The results demonstrate that bubbles tend to form under a smaller laser beam diameter, which is attributed to vortex collisions within the melt pool resulting from a relatively high aspect ratio. As laser heating time increases, the keyhole starts to experience intense fluctuations. During this keyhole mode, the aspect ratio gradually stabilizes, losing its correlation with bubble formation. The influencing parameter shifts to a newly proposed dimensionless parameter, the depth ratio, defined as the ratio of keyhole depth to melt pool depth. This study employs a parameter-morphology-defect correlation analysis to investigate the mechanism of

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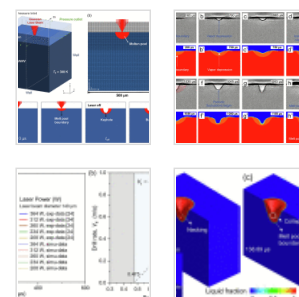
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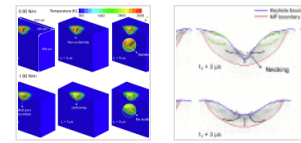
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bubble formation induced by keyhole collapse. Furthermore, it proposes a potential approach for effectively suppressing pore defects by introducing qualitative guidance based on real-time monitoring of melt pool dynamics (aspect ratio, depth ratio) to optimize process parameters.



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Keywords

Bubble formation; Laser melting; Keyhole closure mechanism; Melt pool characteristics; Multi-physics CFD simulation

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1. Introduction

Laser material processing is an advanced manufacturing technology that has been widely applied in processes such as laser welding, surface engineering, additive manufacturing, and laser drilling. Currently, it plays critical roles in aerospace, biomedical, automotive, and electronics industries, and its scope is expected to expand further. The laser material process begins with laser heating the material surface, causing melting and subsequently forming a melt pool (MP). At a high energy density, metallic vapor forms and generates recoil pressure. This pressure pushes down the molten metal, creating a long, slender gas cavity called a vapor depression zone (i.e., keyhole) [1]. Complex physical phenomena affect keyhole formation and dynamics, including phase-change thermodynamics (melting, solidification, and vaporization), multiphase thermo-fluid flow, evaporation-induced recoil pressure, and rapid heating and cooling rates ($\sim 10^4\text{--}10^6\text{ K}\cdot\text{s}^{-1}$) [2], which are challenging to characterize experimentally. Therefore, understanding keyhole dynamics and the associated mechanisms requires high-fidelity numerical models to complement experimental observations [3].

Keyhole-induced porosity is a prevalent defect in laser melting processes (e.g., laser welding and laser powder bed fusion). Although keyhole formation facilitates deeper penetration of laser energy into the material, necking and collapse of the keyhole can trap bubbles within the MP [4]. These bubbles not only significantly reduce energy absorption but may also produce keyhole porosity and deteriorate surface integrity in the processed workpiece [5], [6]. Previous studies have extensively investigated keyhole-induced bubble dynamics [7], revealing complex processes such as keyhole fluctuations [8], collapse, and the subsequent formation [9], [10], growth [11], contraction, and migration of bubbles [12], [13]. The formation of keyhole-induced porosity primarily involves two steps [14]. First, the collapsing keyhole entraps metal vapor and/or shielding gas beneath the free surface, forming bubbles in MP. Second, these bubbles are captured by the solid front, resulting in porosity. Except for converting to porosity, the bubbles have two additional fates: escaping through the free surface or merging with the transiently reopened keyhole [12]. Prevention of pore formation can be achieved by interrupting one of those steps [15]. Extensive research on the fates of bubbles has been carried out, focusing mainly on bubble trapping and escape [16] or on reducing porosity in components through post-processing [17] or remelting [18]. However, investigations into the mechanisms governing the initial bubble formation remain limited. Yet this is precisely the key to reducing porosity at its source, and a thorough investigation of bubble-formation mechanisms is fundamental to mitigating pores in laser-processed materials. Fan et al. [19] demonstrate that applying a horizontally oriented magnetic field suppresses keyhole instability by driving thermoelectric-magnetohydrodynamic flows, thereby inhibiting bubble formation. Ke et al. [20] reveal that an oscillating laser beam generates a

counterclockwise vortex flow pattern that increases the keyhole stability in numerical simulations, thereby aiding in the prevention of a keyhole collapse and bubble formation. Dakka et al. [21] predict keyhole stability that improves when oscillation amplitude exceeds a frequency-dependent threshold, with transitions from unstable to stable regimes observed.

Previous works have related the fate of bubbles to the morphology of MP and keyhole. A comparative study of two laser power levels revealed that pore formation is caused by the excessive depth of the keyhole cavity [22]. Huang et al. [11] discovered a transition regime between the stable and unstable keyhole regimes, where the keyhole morphology changes from wide and shallow in the transition regime to narrow and deep in the unstable keyhole regimes. These studies focus on the transition of keyhole morphology across different stages, effectively correlating bubble formation with the depth or width characteristics of the keyhole. While others have focused on the influence of melt pool morphology. Hayat et al. [23] showed that a circular laser spot concentrates heat to form a deep, narrow keyhole and a correspondingly narrow melt pool with high-speed flow, contributing to the instability of the necking of the keyhole and an increased occurrence of collapse events. Aggarwal et al. [16] emphasized the significant impact of melt pool morphology on bubble formation. A link between a high melt pool aspect ratio and pore formation has been established based on cases with varying laser power and diameter, as well as laser heating time. But its effectiveness under the condition of a single variable still needs to be verified, as conclusions drawn from controlled single-variable studies are more directly applicable to engineering parameter optimization. Furthermore, this study focused on the influence of the aspect ratio during the bubble migration and escape stages, without exploring its role or specific mechanisms in the bubble formation stage, which is critical for suppressing pore generation at its origin. Furthermore, when the aspect ratio gradually stabilizes over time, it remains unclear whether other morphological features strongly correlate with the formation of bubbles. Clarifying this relationship will not only advance the understanding of keyhole collapse and bubble formation mechanisms but also provide practical guidance for optimizing process parameters to suppress pore defects in engineering applications.

To address these limitations, the transient melting process of Ti-6Al-4V under a stationary laser using a physics-based 3D model is simulated, which accounts for laser energy absorption and reflection, heat transfer, phase transformation, and fluid flow. This work systematically investigates the relationship between parameters, morphology, and bubble formation mechanisms through independent control of parameters. Furthermore, the depth ratio (keyhole depth/MP depth) is introduced to provide an effective preliminary stability criterion, particularly in scenarios where the aspect ratio (MP depth/MP width) tends to stabilize over time. The modeling results reproduced the morphology of MP and keyhole observed in the experiments by Cunningham et al. [24]. The depth-time curves also show good agreement with the experimental results, thereby validating our physics-based modeling approach. We compared the keyhole closure phenomenon and bubble formation for different materials and laser parameters, including surface tension, laser beam diameter, and laser exposure time. To facilitate comparison, the laser was turned off to eliminate the potential for merging with the newborn keyhole. Thus, our research focuses on the molten pool dynamics and keyhole closure behavior during the laser-off phase. This study aims to reduce defect formation from its origin by providing direction for the implementation of in-situ monitoring of melt pool morphology [25].

2. Computational model

The morphological evolution of the molten pool and keyhole during stationary laser melting of Ti-6Al-4V bare plates was simulated using a finite volume method-based computational model. The model captures the complex thermo-fluidic phenomena occurring in the transient MP, including phase changes, Marangoni flow, buoyancy force, vaporization-induced recoil pressure, reflections of the irradiating laser beam on the vapor depression surface, and heat losses (convection, radiation, and evaporation). We aim to elucidate the fundamental mechanisms governing keyhole closure and gas bubble formation during the laser-off phase. A set of conservation equations for mass, momentum, and thermal energy, along with the volume-of-fluid equation, has been solved to calculate the temperature field and velocity field, thereby precisely capturing the morphology of MP. Numerical simulations in this study were conducted using the commercial software ANSYS Fluent. Complex physical processes were modeled via User-Defined Functions (UDFs) that implemented the source terms of the governing equations. To balance solution accuracy and computational efficiency, several assumptions commonly used in the interaction of laser and metal processes are adopted. (1) The MP is in a laminar flow state and incompressible with Newtonian viscosity [26]; (2) the initial temperature throughout the computational domain is set at 300K; (3) the physics of the vapor plume ejecting from the MP is not explicitly accounted for. Instead, source terms in the momentum and energy conservation equations are used to account for the influence of vaporization.

2.1. Governing equations

The governing equation of mass conservation is shown as:

$$\frac{\partial \bar{\rho}}{\partial t} + \nabla \cdot (\bar{\rho} \vec{v}) = 0 \quad (1)$$

where the mixture density is calculated by: $\bar{\rho} = \alpha_1 \rho_1 + (1 - \alpha_1) \rho_2$ [kg/m³], t is the flow time [s], and $\vec{v}$ is the mass-averaged velocity $\vec{v} = [\alpha_1 \rho_1 \vec{v}_1 + (1 - \alpha_1) \rho_2 \vec{v}_2] / \bar{\rho}$ [m/s].

The Volume of Fluid (VOF) Model is employed to track the liquid–gas interface, and the volume fraction equation is shown as:

$$\frac{\partial \alpha_1}{\partial t} + \nabla \cdot (\vec{v} \alpha_1) = 0 \quad (2)$$

where α_1 represents the volume fraction of the metallic phase. $\alpha_1 = 0$ corresponds to a cell completely occupied by the gaseous phase, while $\alpha_1 = 1$ indicates a cell that is full of metal, and cells with $0 < \alpha_1 < 1$ are the free surfaces. In the continuum formulation model, the weighted volume-averaged thermo-physical properties are calculated using the classical mixture theory. For any thermo-physical property $\bar{\Psi}$, the relation is

$$\bar{\Psi} = \alpha_1 \Psi_1 + (1 - \alpha_1) \Psi_2, \quad \Psi \in [\rho, \mu, C_p, k] \quad (3)$$

where ρ is density, μ is dynamic viscosity, C_p is specific heat capacity, and k is thermal conductivity. Ψ_1 (Ti6Al4V) and Ψ_2 (Argon gas) are the properties of the immiscible phases.

The Navier-Stokes momentum conservation equation is given by:

$$\frac{\partial (\bar{\rho} \vec{v})}{\partial t} + \nabla \cdot \bar{\rho} \vec{v} \otimes \vec{v} = -\nabla P + \nabla \cdot \left(\mu \left(\nabla \cdot \vec{v} + (\nabla \cdot \vec{v})^T \right) \right) + \bar{\rho} \vec{g} + \vec{S}_m \quad (4)$$

where P is the pressure [Pa], $\vec{g}$ represents the gravity vector [m/s], and μ is the dynamic viscosity [Pa·s]. $[\text{kg}/(\text{m}^3 \cdot \text{s})]$ is the momentum source term, including the buoyancy force $\vec{F}_{\text{Buoyancy}}$ [N/m³], Darcy's drag force for the mushy zone $\vec{F}_{\text{Mushy-Drag}}$ [N/m³], the surface tension $\vec{F}_{\text{surface}}$ [N/m³], the Marangoni force $\vec{F}_{\text{Marangoni}}$, and recoil pressure $\vec{F}_{\text{recoil}}$ [N/m³]

expressed as [27].

$$\vec{S}_m = \vec{F}_{Buoyancy} + \vec{F}_{Mushy-Drag} + \vec{F}_{Surface} + \vec{F}_{Marangoni} + \vec{F}_{recoil} \quad (5)$$

In terms of buoyancy force $\vec{F}_{Buoyancy} = -\vec{\rho}(T - T_r)\vec{g}$, β is the thermal expansion coefficient [K^{-1}], T and T_r are the local temperature and reference temperature, respectively [K]. In terms of Darcy's drag force for the mushy zone $\vec{F}_{Mush-Drag} = [(1 - \alpha_1)^2 / (\alpha_1^3 + b)] A_{mush} \vec{v}$, A_{mush} is the mushy zone constant [$kg/m^2 \cdot s$], which quantifies the damping amplitude. A higher value results in a more dramatic reduction of material velocity to zero upon solidification. Excessively high values, however, can introduce numerical instabilities and oscillations [28]. The results demonstrate that a value of $A_{mush} = 2.1 \times 10^7$ ensures a stable calculation process while producing a distinct solidification front that aligns with physical expectations. b is a small number to prevent division by zero. Fundamentally, both Eq. (6) and Eq. (7) originate from the influence of surface tension: Eq. (6) describes the direct effect of surface tension acting normal to the surface, which is often regarded as equivalent to an external pressure, whereas Eq. (7) represents the spatial gradient of surface tension acting parallel to the surface, known as the Marangoni effect.

$$\vec{F}_{Surface} = \kappa \left[\frac{\partial \gamma}{\partial T} (T - T_l) + \gamma_0 \right] \frac{2\bar{\rho}}{\rho_1 + \rho_2} |\nabla \alpha_1| \vec{n} \quad (6)$$

$$\vec{F}_{Marangoni} = \frac{\partial \gamma}{\partial T} [\nabla T - (\vec{n} \cdot \nabla T) \vec{n}] \frac{2\bar{\rho}}{\rho_1 + \rho_2} |\nabla \alpha_1| \quad (7)$$

$$\vec{F}_{recoil} = 0.54 P_0 \exp \left[\frac{L_v M (T - T_v)}{R T T_v} \right] \frac{2\bar{\rho}}{\rho_1 + \rho_2} |\nabla \alpha_1| \vec{n} \quad (8)$$

$$\vec{n} = \frac{\nabla \alpha_1}{|\nabla \alpha_1|} \quad (9)$$

The continuum surface force (CSF) model [29] is proposed to transfer the surface forces to volumetric forces by using a multiplier $[2\bar{\rho}/(\rho_1 + \rho_2)]|\nabla \alpha_1|$, in which ρ_1 and ρ_2 are the densities of metal and gas, respectively [kg/m^3]. Here κ [m^{-1}] is the surface curvature. γ and γ_0 are the surface tension coefficients at the local temperature and liquidus temperature, respectively [N/m]. $\vec{n}$ is the unit normal vector of the liquid–gas surface.

The basic equation for energy conservation can be expressed as:

$$\bar{\rho} \frac{\partial h}{\partial t} + \bar{\rho} (\vec{v} \cdot \nabla h) = \nabla \cdot (k \nabla T) + S_h \quad (10)$$

where h is enthalpy [J/kg], S_h is the energy source term [W/m³], given by:

$$S_h = Q_{laser} + Q_{evaporation} + Q_{convection} + Q_{radiation} \quad (11)$$

During the laser-material process, a laser beam is used as a thermal energy source to induce the fusion of the material. Considering that the laser radiation penetrates into the substrate by reflection, we chose a volumetric heat source with a Gaussian distribution instead of a surface heat source. The heat source does not directly solve complex dynamic physical processes such as keyhole wall reflection, plasma/vapor shielding, and scattering. Instead, it equivalently simulates their combined effects through a rationally designed heat source function with a specific spatial distribution. The distribution of the laser energy intensity followed nearly a Gaussian relationship [30], which was mathematically presented as:

$$Q_{laser} = \frac{3c_s P A}{\pi H \left(1 - \frac{1}{e^3}\right)} \exp \left[\frac{-3c_s}{\log\left(\frac{H}{z}\right)} \left((x - x_0)^2 + (y - y_0)^2 \right) \right] \quad (12)$$

where c_s is the shape coefficient, equal to $3/R_0^2$, R_0 is the mouth diameter of the heat source, and P is the laser power. The evaporation heat loss, $Q_{evaporation}$, is defined as [31]:

$$Q_{evaporation} = -0.82 \frac{L_v M}{\sqrt{2\pi M R}} P_0 \exp \left[\frac{L_v M (T - T_v)}{R T T_v} \right] \frac{2\bar{p}}{\rho_1 + \rho_2} |\nabla \alpha_1| \quad (13)$$

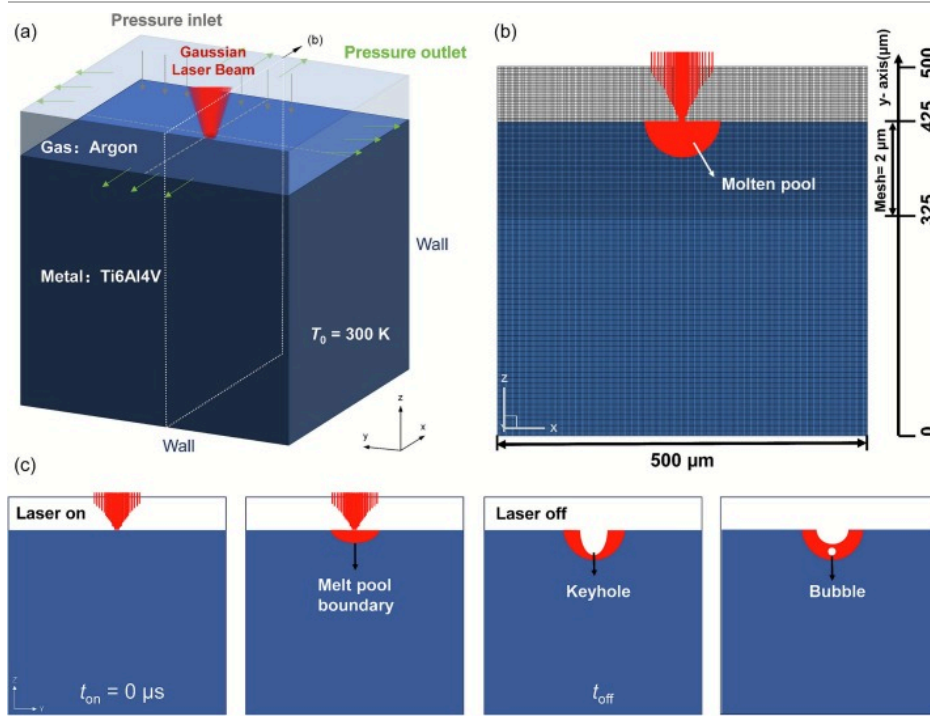
The heat loss due to convection $Q_{convection}$ and radiation $Q_{radiation}$ [W/m³] is given by [32]:

$$Q_{convection} = -h_c (T - T_0) \frac{2\bar{p}}{\rho_1 + \rho_2} |\nabla \alpha_1| \quad (14)$$

$$Q_{radiation} = -\sigma \epsilon (T^4 - T_0^4) \frac{2\bar{p}}{\rho_1 + \rho_2} |\nabla \alpha_1| \quad (15)$$

2.2. Numerical simulation and model validation

Fig. 1a depicts the three-dimensional computational domain employed in the present study: a Ti-6Al-4V substrate of dimension (500μm×500μm×425μm), surmounted by an argon-filled gas layer of dimension (500μm×500μm×75μm). The laser is continuously irradiated at the center of the Ti-6Al-4V surface under an ambient temperature of $T_0=300$ K. To balance computational efficiency with accuracy, the domain is discretized via a graded mesh with a minimum size of 2μm, while the melting-affected zone (325μm<z<425μm) is resolved with a uniform 2μm×2 μm×2 μm element size to ensure high fidelity, as shown in Fig. 1b (y=250μm). Numerical simulations are conducted with an adaptive time step varying from 5×10^{-8} to 2×10^{-7} s. The chosen timestep and grid size yield stable, consistent results with high-resolution keyhole and melt-pool profiles. The boundary conditions were applied as follows: the +Z direction was set as a pressure boundary, and the -Z direction was defined as a wall boundary. For the ±Y and ±X directions, the boundary conditions differed between the gas domain (z>425μm) and the metal domain (z<425μm). In the gas domain, all boundaries were treated as walls, while in the metal domain, all boundaries were set as pressure boundaries. Table 1 lists properties of Ti-6Al-4V used in the simulations. At the initial instant, a stationary laser is switched on ($t_{on}=0$) and applied to the material surface. The formation of MP occurs when the surface temperature exceeds the liquidus temperature ($T_l=1923$ K). Upon reaching the vaporization temperature ($T_v=3313$ K), intense metallic vapor generates recoil pressure that depresses the melt surface into a long, slender keyhole, as shown in Fig. 1b. A schematic diagram of the processing sequence is presented in Fig. 1c. After the laser is shut off at t_{off} , the transient response of the MP and the underlying mechanism of bubble formation are investigated. Parameters used in the simulations for cases in this work are listed in Table 2.



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Fig. 1. Computational domain and mesh. (a) The three-dimensional computational domain with boundary conditions. (b) Meshing of the cross-section at $y=250\mu\text{m}$. (c) Schematic diagram of the processing procedure in the simulation.

Table 1. Thermal-physical parameters of Ti-6Al-4V in the simulation.

Physical properties of Ti-6Al-4V	Symbol	Value
Density of solid	ρ_s	4430 kg/m ³
Density of liquid	ρ_l	4122 kg/m ³
Dynamic viscosity	μ	0.00236 Pa·s
Solidus temperature	T_s	1877 K
Liquidus temperature	T_l	1923 K
Vaporization temperature	T_v	3313 K [33]
Surface tension at the liquidus temperature	γ_0	1.65 N/m [16]
Temperature coefficient of surface tension	$\partial\gamma/\partial T$	-2.4×10^{-4} N/(m·K) [16]
Specific heat capacity of solid	$C_{p,s}$	780 J/(kg·K) [34]
Specific heat capacity of liquid	$C_{p,l}$	930 J/(kg·K) [34]
Thermal conductivity of solid	k_s	13 W/(m·K) [34]
Thermal conductivity of liquid	k_l	48 W/(m·K) [35]
Latent heat of fusion	L_f	2.85×10^5 J/kg [33]
Latent heat of vaporization	L_v	8.88×10^6 J/kg [33]
Convection coefficient	h_c	25 W/m ² ·K [8]
Gravity	g	9.81 m/s ²
Ambient pressure	P_0	101325 Pa

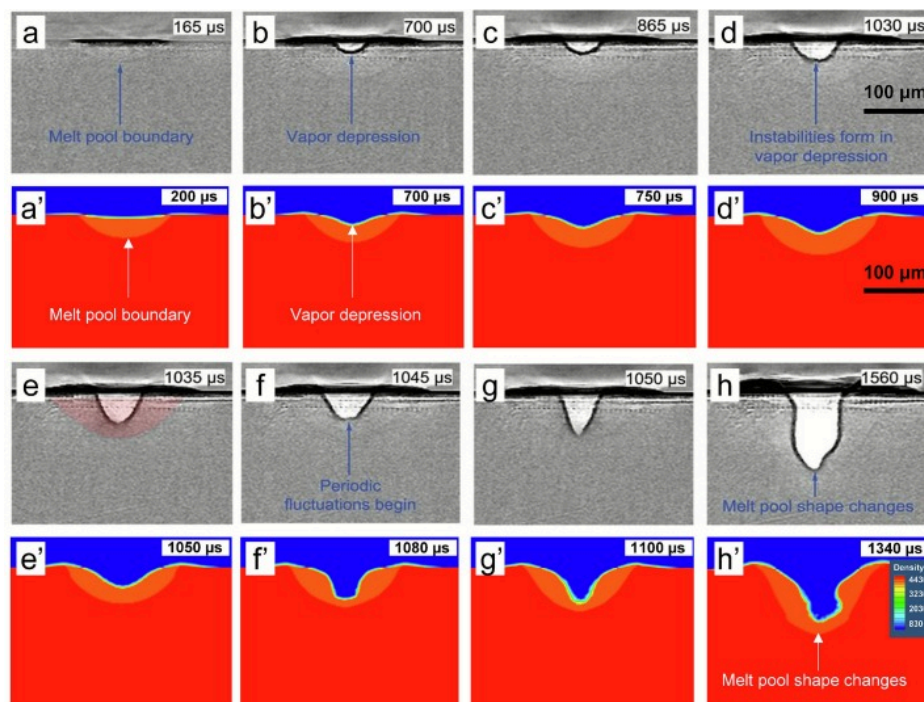
Physical properties of Ti-6Al-4V	Symbol	Value
Ambient temperature	T_0	300K
Molar mass	M	0.04658kg/mol [33]
Radiation emissivity	ε	0.18 [8]
Mushy zone parameter	A_{mush}	$2.1 \times 10^7 \text{ kg}/(\text{m}^2 \cdot \text{s})$
Laser absorption coefficient	A	0.3 [7]
Stefan-Boltzmann constant	σ	$5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \cdot \text{K}^4)$

Table 2. Parameters used in simulations. The laser power is 312W.

Case ID	Surface tension γ_0 (N/m)	Parameter α in recoil pressure	Laser beam diameter (μm)	Laser exposure time t_{off} (μs)
A0	1.65	0.54	140	135
A1	0.90	0.54	140	65 (t_1)
A2	1.80	0.54	140	65
A3	1.35	0.54	140	70 (t_2)
A4	1.20	0.54	140	70
A5	1.20	0.40	140	70
B1	1.65	0.54	130	75 (t_3)
B2	1.65	0.54	140	75
B3	1.65	0.54	150	75
C1	1.90	0.54	140	90
C2	1.90	0.54	140	105
C3	1.90	0.54	140	125
C4	1.90	0.54	140	130

The model predictions are first benchmarked against the representative experiment of Cunningham et al. [12], in which a stationary 156W, 140 μm -diameter laser was directed vertically onto a bare Ti-6Al-4V plate. Snapshots of the ensuing melt-pool formation and evolution are presented in Fig. 2: a-h show the experimental observations from [12], while a'-h' display the corresponding simulation results with density contours. Under continuous irradiation by a stationary laser beam, the melt-pool morphology evolves through two distinct stages: an initial hemispherical (conduction mode) phase with a quasi-semicircular profile (Fig. 2a'-e'), followed by a transition to a bimodal (keyhole mode) configuration characterized by a bowl-shaped upper region surmounting a central spike at the bottom (Fig. 2f'-h'). This transition point can be directly visualized as a clear turning point in Fig. 3a. The keyhole depth at this transition point is defined as the critical depth, and the corresponding time of this transition is defined as the transition time. In the conduction mode, the simulated melt pool depth is slightly greater than the experimental value, which is attributed to the pre-deposition of energy inside the material by the Gaussian rotating volumetric heat source. After keyhole evolution begins, both the width and depth of the simulated melt pool become slightly smaller than the measured values. This discrepancy stems primarily from the simplified treatment of the energy absorption mechanism within the keyhole. Although the employed volumetric

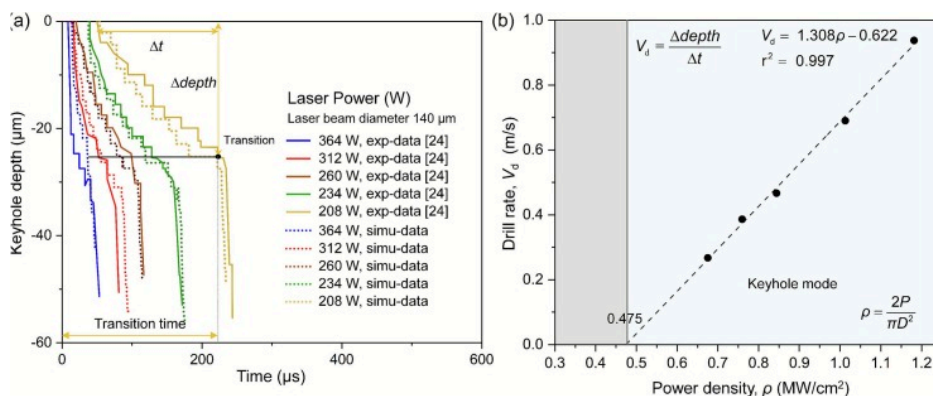
heat source reasonably captures the three-dimensional spatial distribution of energy, it fails to adequately represent the enhancement in absorptivity caused by multiple reflection effects. The resultant relative deficit in energy input consequently leads to an inadequately developed melt pool. The deviation in simulated keyhole morphology is correlated with this inaccuracy in melt pool dimensions.



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Fig. 2. The evolution of keyhole and MP of Ti-6Al-4V alloy under a stationary laser beam (Laser power=156W, beam spot size=140μm). (a–h) In-situ ultrahigh speed X-ray imaging results [24]. Light gray regions denote the MP, and the white regions denote the vapor depression or keyhole. (a'–h') Longitudinal section of the simulation results with density contours.



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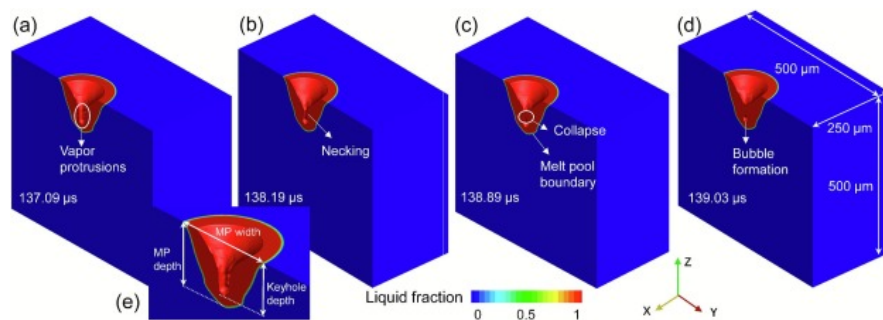
Fig. 3. Keyhole drilling under stationary laser illumination. (Laser beam diameter=140μm) (a) Keyhole depth evolution of Ti-6Al-4V alloy during stationary laser melting. (b) Drill rate of the laser as a function of power density before the transition. The black dashed line is the linear fit to the drill rates.

For a systematic evaluation and enhancement of the model's accuracy, this study compares the errors between numerical simulations and experimental results [24] across five different power levels. Fig. 3a presents the keyhole depth as a function of laser exposure time at different laser powers for a fixed beam diameter of 140 μm . The model agrees best with the experimental measurements at an intermediate power of 312W. Hence, all subsequent simulations are performed at this power. As shown in Fig. 3a, the critical depth remains essentially constant at about 25 μm across all tested laser powers for a given spot size. The ratio of the critical depth to the time gives the drilling rate, $V_d = \Delta \text{depth} / \Delta t$ (m/s), which increases linearly with the laser power density, $\rho = 2P / \pi D^2$ (MW/cm²), as shown in Fig. 3b. Consequently, the transition time decreases sharply with increasing laser power. If the laser power is sufficiently high (364W), the drilling rate becomes nearly identical to the continuous deepening rate after keyhole instability, causing the turning point in depth-time curves (transition signature) to disappear. If the laser power is sufficiently low, melting remains permanently in conduction mode without transitioning to keyhole mode due to insufficient power density. Based on this, a material-dependent critical power density is defined, over which the keyhole mode is active. As shown in Fig. 3b, the drill rate in our simulation followed approximately a linear relation ($r^2=0.997$) with power density after the critical threshold of 0.475MW/cm², consistent with previous estimates [24].

3. Results and discussion

3.1. Bubble formation mechanism after the transition

During the stationary laser irradiation of the keyhole mode, the phenomenon of bubble formation is observed for case A0. As illustrated in Fig. 4a–d, four stages of bubble formation are sequentially demonstrated: vapor protrusion, keyhole necking, keyhole collapse, and bubble formation, respectively. The entire process is completed within microseconds ($\sim 2 \mu\text{s}$). Fig. 4e visually characterizes three key morphological parameters. (1) MP depth: the maximum vertical distance from the substrate surface to the molten zone's deepest point, reflecting energy penetration. (2) Keyhole depth: measured from the surface to the vapor cavity tip, indicating vapor recoil pressure limits. (3) MP width: the molten zone's maximum lateral span at the surface, representing laser-material interaction extent.



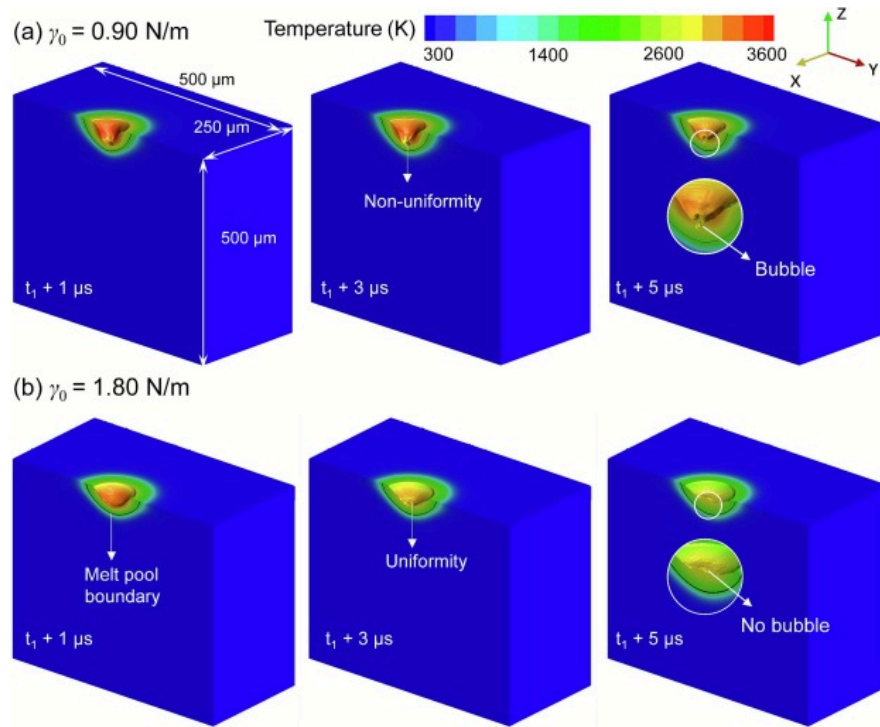
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Fig. 4. The formation process of a bubble for case A0. (Laser power=312W, laser beam diameter=140 μm , $t_{\text{off}}=135 \mu\text{s}$). (a–d) Evolution of the melt-pool morphology at half of the simulation domain. (e) Definition schematics of morphological parameters for the MP. All depths are measured from the original substrate surface.

As shown in Fig. 5, the bubble formation dynamics with temperature field for cases A1 ($\gamma_0=0.90\text{N/m}$) and A2 ($\gamma_0=1.80\text{N/m}$) under constant laser parameters is investigated. At

$t_1 + 1\mu\text{s}$ for both case A1 and case A2, the temperature on the MP surface reaches a maximum of $\sim 3500\text{K}$ at the center and decreases monotonically along the radial direction toward the periphery. The difference lies in that the MP surface with $\gamma_0 = 0.90\text{N/m}$ exhibits significant temperature non-uniformity (Fig. 5a) but exhibits uniformity when $\gamma_0 = 1.80\text{N/m}$ (Fig. 5b). When $\gamma_0 = 0.90\text{N/m}$, there is an overall decrease in the temperature, accompanied by the necking occurring at $t_1 + 3\mu\text{s}$, as shown in Fig. 5a. Subsequently, a bubble is observed forming directly below the keyhole at $t_1 + 5\mu\text{s}$. In contrast, when $\gamma_0 = 1.80\text{N/m}$, the MP surface contracts smoothly upward at $t_1 + 3\mu\text{s}$, resulting in the complete lifting of the gas above it by $t_1 + 5\mu\text{s}$, as shown in Fig. 5b. Simulation results reveal that the bubble-formation behavior is markedly different depending on the surface-tension value, with elevated surface tension significantly suppressing the formation of bubbles.



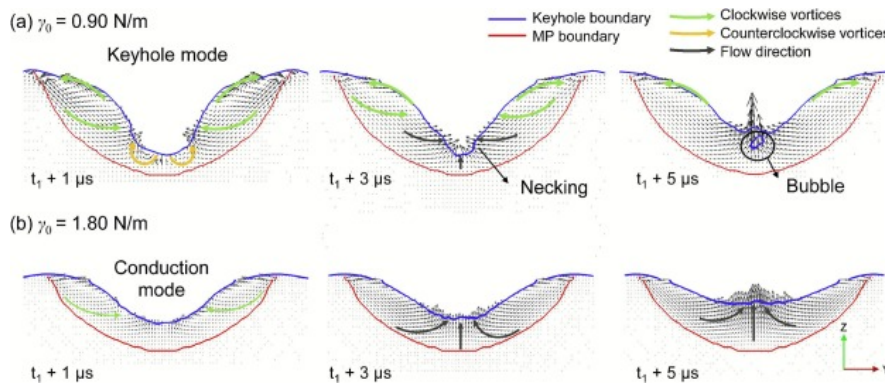
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Fig. 5. The keyhole morphology and temperature field at one half of the simulation domain for cases A1 and A2. (a) A1: $\gamma_0 = 0.90\text{N/m}$, (b) A2: $\gamma_0 = 1.80\text{N/m}$. (Laser power = 312W, laser beam diameter = 140 μm , $t_{\text{off}} = t_1\mu\text{s}$).

To elucidate the mechanism of the difference in bubble formation for cases A1, A2, the velocity field on the cross-section $x = 250\mu\text{m}$ is presented in Fig. 6. It is found that at $t_1 = 65\mu\text{s}$, the melt pool in case A1 is in keyhole mode, exhibiting a spike-shaped profile (Fig. 6a). In contrast, case A2 remains in conduction mode with quasi-semicircular morphology (Fig. 6b). Thus, they have distinctive velocity field patterns, consequently inducing different physical phenomena. As shown in Fig. 6, Fig. 6, a center-to-periphery flow driven by the Marangoni force is established on the MP surface due to the temperature gradient. The flow is bounced back when it hits the unmelted solid substrate, and a counterclockwise vortex (green arrows) is formed near the solid-liquid boundaries. This bounced-back flow influences the balance between surface tension and recoil pressure of the surface, causing disturbances within the MP. This will lead to the asymmetry and instability in the keyhole morphology [8]. Concurrently, a clockwise vortex (yellow arrows) is formed at the MP bottom. This vortex originates from the recirculating current that balances the downward jet produced by the recoil pressure.

These two vortices generate an impinging jet that strikes the mid-section of the keyhole wall. As the surface tension is relatively low (A1: $\gamma_0=0.90\text{N/m}$), the melt-pool surface retracts upward slowly, allowing the impinging jet to induce the collapse of the keyhole, forming gas bubbles. However, with increased surface tension (A2: $\gamma_0=1.80\text{N/m}$), vortices within the MP are markedly attenuated as shown in Fig. 6b. As mathematically described by the surface tension model in Eq., increasing the surface tension at the liquidus temperature γ_0 while holding the temperature coefficient $\partial\gamma/\partial T$ constant elevates the surface tension γ at any given temperature. The enhanced surface tension affects the melt pool flow field through two primary mechanisms. First, it suppresses vapor depression, yielding a smaller interface curvature as illustrated at $t_1 \mu\text{s}$ in Fig. 6b. According to Eq.(7), the smaller curvature attenuates the Marangoni force, resulting in a weaker green vortex at $t_1+1\mu\text{s}$ in Fig. 6b. Second, the enhanced surface tension quickly attains dominance in the competitive balance at $t_1+1\mu\text{s}$, the velocity of flows is smaller, and the keyhole is more stable, as evidenced at $t_1+3\mu\text{s}$. Therefore, when the surface tension is increased to 1.80N/m , the melting remains in conduction mode at $t_1 \mu\text{s}$, with the melt pool exhibiting significantly lower flow velocities. This low-velocity flow field stabilizes the keyhole and effectively prevents bubble formation, as evidenced by the absence of bubble cavities at $t_1+5\mu\text{s}$ Fig. 6b). A comparison between Fig. 6a and b demonstrates that increasing the surface tension shifts the melt pool into conduction mode at time $t_1 \mu\text{s}$, resulting from the delayed transition, which effectively suppresses bubble formation.

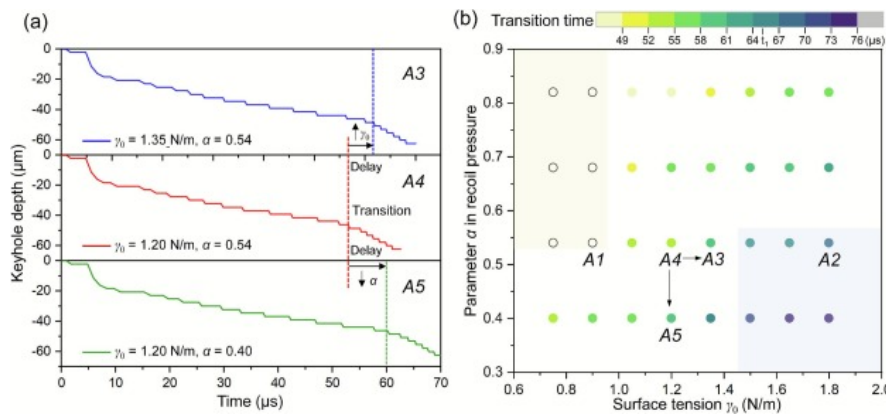


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Fig. 6. Velocity field in the cross-section $x=250\mu\text{m}$ for cases A1 and A2. (a) A1: $\gamma_0=0.90\text{N/m}$, (b) A2: $\gamma_0=1.80\text{N/m}$. (Laser power= 312W , laser beam diameter= $140\mu\text{m}$, $t_{\text{off}}=t_1 \mu\text{s}$). The red line (liquid fraction= 0.5) indicates the liquid–solid interface, while the blue line ($\alpha_1=0.5$) marks the keyhole boundary. The clockwise vortices that flow from the bottom to the top of the MP, driven by the recoil pressure, are indicated by the yellow arrow. The counterclockwise vortices that flow from the central region of the MP to the periphery, driven by Marangoni convection, are indicated by the green arrow. The key flow direction, which is directly related to the difference in bubble formation phenomena, is indicated by the black arrow.

The impact of varying surface tension γ_0 and recoil pressure parameter α on the transition time was compared by cases A3–A5, as shown in Fig. 7a. It demonstrates that reduced recoil pressure (A4 vs A5) or increased surface tension γ_0 (A4 vs A3) both retard keyhole deepening and delay the transition, as shown in Fig. 7a. It can be determined that enhancing surface tension or diminishing recoil pressure consistently delay the transition thereby effectively suppresses bubble formation.



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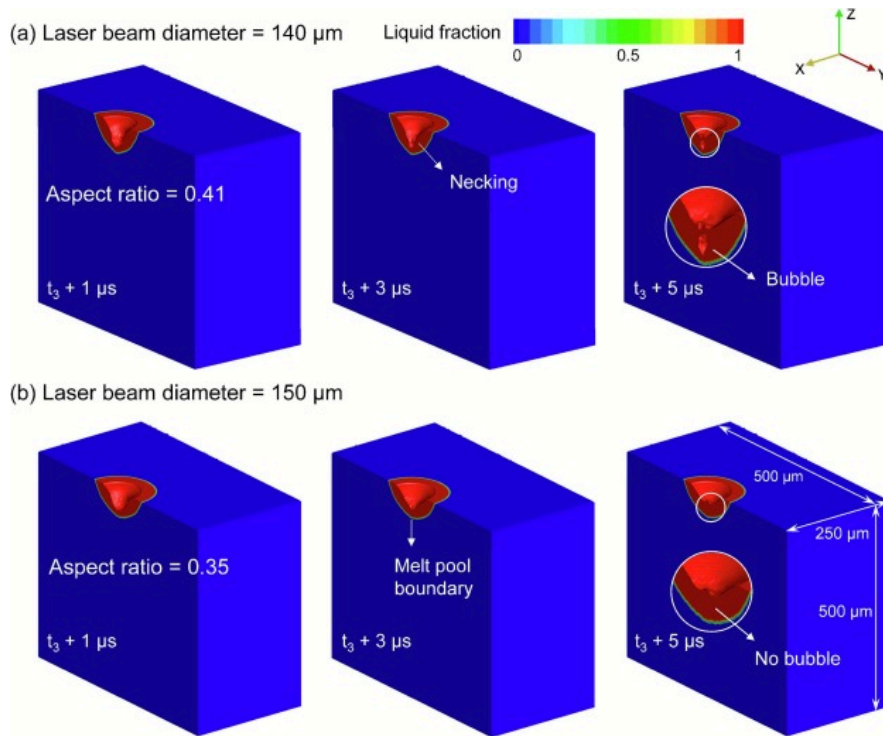
Fig. 7. Variation of transition time with surface tension γ_0 and parameter α in recoil pressure (Laser power=312W, laser beam diameter=140 μ m). (a) Keyhole depth over time with different γ_0 and α ($t_{\text{off}}=t_2$ μ s): case A3: $\gamma_0=1.35$ N/m, $\alpha=0.54$, case A4: $\gamma_0=1.20$ N/m, $\alpha=0.54$, and case A5: $\gamma_0=1.20$ N/m, $\alpha=0.40$. (b) The distribution of transition time in the parameter space across surface tension γ_0 and recoil pressure parameter α .

To gain deeper insights into the interplay of surface tension and recoil pressure, the transition time across expanded parameter ranges is systematically quantified. As shown in Fig. 7b, the color intensity of solid points represents the magnitude of the transition time from 49 to 76 μ s, while the hollow points denote cases where no transition signature is observed in depth-time curves. It can be seen that the transition time increases with decreasing α or increasing γ_0 , demonstrating a positive correlation with surface tension but an inverse dependence on recoil pressure. This originates from dynamically balanced competition between recoil pressure and surface tension during keyhole evolution. Recoil pressure facilitates keyhole formation while surface tension opposes it. When recoil pressure dominates, it depresses the MP free surface, thereby increasing keyhole depth; conversely, surface tension dominance elevates the free surface, reducing depth. Under extreme conditions ($\gamma_0 < 0.90$ and $\alpha > 0.54$), as marked by the hollow points, recoil pressure dominates overwhelmingly. This results in indistinguishable drill rates between conduction mode and keyhole mode, thereby eliminating the transition signature in depth-time curves. The parameter range with this characteristic is denoted by the shallow yellow zone (e.g., case A1). The light blue zone indicates parameter ranges where the transition time exceeds t_1 , corresponding to cases (e.g., case A2) remaining in conduction mode at t_1 μ s. Moreover, when $\alpha=0.4$, increasing surface tension from 1.05 N/m to 1.80 N/m extends the transition time from 57.0 μ s to 74.6 μ s (a 31% increase). This effect decreases to 29% at $\alpha=0.54$ and further reduces to 25% at $\alpha=0.82$. This demonstrates that higher recoil pressure can mitigate the prolonging effect of increased surface tension on the transition time, thereby offering a direct and quantitative measure of the competitive relationship between surface tension and recoil pressure.

3.2. Influence of laser beam diameter on aspect ratio and bubble formation

We investigated the phenomenon of bubble formation under laser beam diameters of 140 μ m and 150 μ m. Fig. 8 illustrates the morphologies of the MP and keyhole after the laser is turned off at t_3 μ s. The molten region (red) corresponds to a liquid fraction of 1.0, while the completely unmelted zone (blue) has a liquid fraction of 0. The intermediate areas with liquid fractions between 0 and 1 represent the mushy zone. The evolution of the keyhole profile and different bubble formation phenomena can be observed within the molten pool. As shown in Fig. 8a, when the diameter was 140 μ m, necking was

observed on the sidewall at the bottom of the keyhole at $t_3 + 3\mu\text{s}$, followed by necking and the formation of an instantaneous bubble at $t_3 + 5\mu\text{s}$. However, when the diameter was $150\mu\text{m}$, these phenomena were not observed. Instead, the MP maintained a relatively stable morphology after the laser was turned off. Subsequently, the MP free surface rises without forming any bubbles, as shown in Fig. 8b. The study revealed that MP under a smaller-diameter beam exhibited a higher aspect ratio, after which bubble formation was observed. In contrast, MP under a larger-diameter beam showed a lower aspect ratio with no bubble generation. This phenomenon suggests a potential correlation between the melt pool aspect ratio and the bubble formation mechanism.

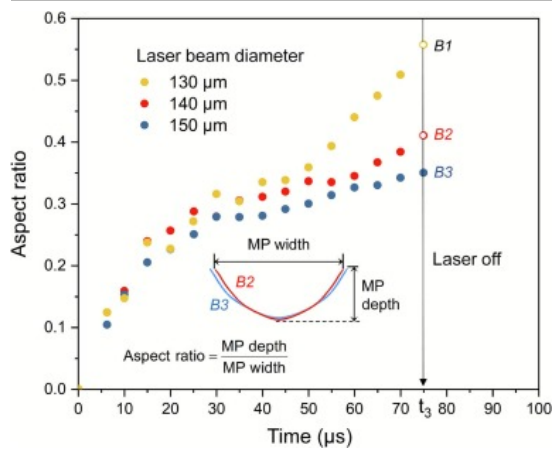


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Fig. 8. The keyhole morphology and liquid fraction distribution at one half of the simulation domain for cases of different laser beam diameters for cases B2 and B3. The molten region (red, liquid fraction=1.0), the completely unmelted zone (blue, liquid fraction=0), and the mushy zone (liquid fraction between 0 and 1) are shown. The evolution of the keyhole profile and bubble formation phenomena within the molten pool is observed. (a) B2: Laser beam diameter= $140\mu\text{m}$, (b) B3: Laser beam diameter= $150\mu\text{m}$ (Laser power= 312W , $t_{\text{off}}=t_3\mu\text{s}$).

Fig. 9 depicts the temporal evolution of the aspect ratio (MP depth/MP width) for laser beam diameters of $130\mu\text{m}$ (B1), $140\mu\text{m}$ (B2), and $150\mu\text{m}$ (B3) under constant laser power (312W) and laser exposure time ($t_3\mu\text{s}$). As shown, the aspect ratio exhibits a nonlinear increase with time, consistent with the experimental observations of [24]. It can be seen that a smaller laser diameter has a higher growth rate of aspect ratio, thereby the smallest diameter yielding the highest aspect ratio at $t_3\mu\text{s}$ (case B1). This occurs because a smaller beam narrows the melt pool by minimizing the heat-affected zone while increasing penetration depth due to higher energy density, ultimately forming a deep and narrow melt pool. Consequently, the MP generated by the $140\mu\text{m}$ laser exhibits a narrow and deep profile at $t_3\mu\text{s}$ (red line for case B2), which can be visualized from the morphology diagrams in Fig. 9. Therefore, it has a higher aspect ratio. In contrast, the $150\mu\text{m}$ laser forms a wide and shallow MP (blue line for case B3), resulting in a smaller aspect ratio.

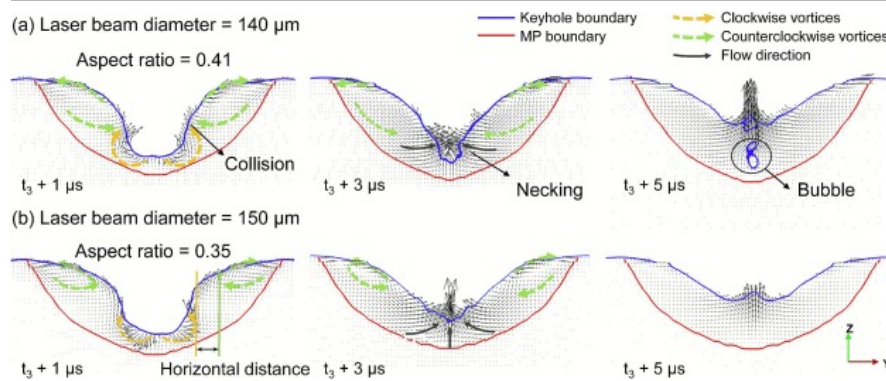


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Fig. 9. The aspect ratio versus time for cases *B1*, *B2*, and *B3* with different laser beam diameters. Hollow markers denote bubble formation upon laser off at the designated time point ($t_3=75\mu\text{s}$), whereas solid markers represent bubble-free conditions (Laser power=312W, $t_{\text{off}}=t_3\mu\text{s}$).

Fig. 10 shows the velocity field map at the melt pool cross-section $x=250\mu\text{m}$ with MP (red line) and keyhole boundary (blue line) for cases *B2*, *B3* to further investigate the mechanism linking aspect ratio and bubble formation phenomena. Research shows that a high aspect ratio can provide favorable conditions for vortex collision, resulting in keyhole necking and subsequent bubble formation. As shown in Fig. 10a, under a laser beam diameter of $140\mu\text{m}$, a melt pool with a high aspect ratio is typically narrow and deep. This geometry prevents any horizontal separation between two vortices, leading to their collision. One is a clockwise vortex that flows from the bottom to the top of the MP, indicated by the yellow arrow; the other is a counterclockwise vortex that flows from the central region of the MP to the periphery, driven by Marangoni convection, indicated by the green arrow. Due to the constrained width of the MP, these two counter-rotating vortices collide and simultaneously impact the middle section of the keyhole wall at $t_3+1\mu\text{s}$. The vortex collision causes necking formation at $t_3+3\mu\text{s}$. As shown, radially inward high-velocity flow (indicated by the black arrow) at the MP surface and insufficient upward velocity in the central region induce keyhole collapse, leading to a bubble formed at $t_3+5\mu\text{s}$. Thus, the high aspect ratio (0.41) MP promotes the collision of two vortices due to its narrow-deep characteristics, promoting bubble formation. For a laser beam diameter of $150\mu\text{m}$, the vortices can also be observed in the MP at $t_3+1\mu\text{s}$ (Fig. 10b). As shown, the wide and shallow MP provides sufficient horizontal distance for the vortices, preventing their collision. Instead, the clockwise vortex (yellow arrows) at the bottom upward parallel to the keyhole wall, and merges into the counterclockwise flow (green arrows). Subsequently, the keyhole evolves into a conical morphology driven by surface tension at $t_3+3\mu\text{s}$. This conical keyhole undergoes surface-tension-driven closure synchronized with free surface uplift, achieving a bubble-free condition at $t_3+5\mu\text{s}$.



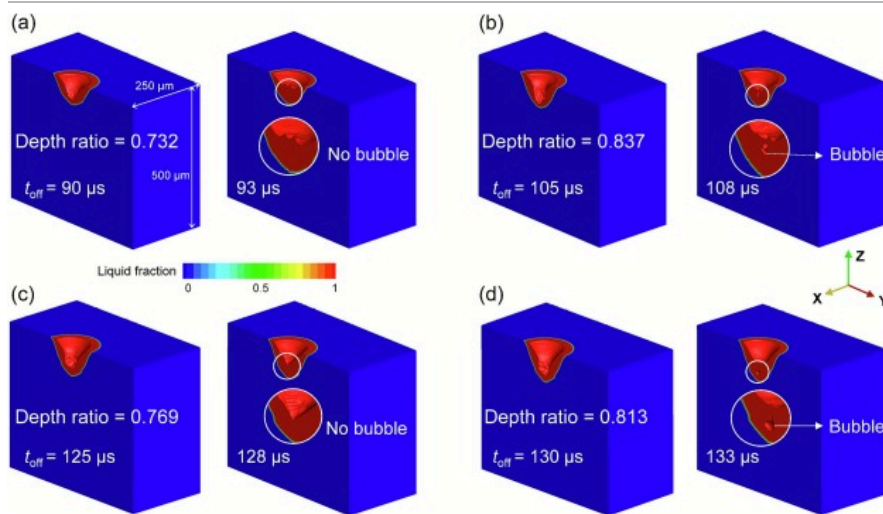
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Fig. 10. Velocity field in the cross-section $x=250\mu\text{m}$ for cases *B2*, *B3*. (a) *B2*: Laser beam diameter= $140\mu\text{m}$. (b) *B3*: Laser beam diameter= $150\mu\text{m}$. The red line (liquid fraction=0.5) indicates the liquid–solid interface, while the blue line ($\alpha_1=0.5$) marks the keyhole boundary. The clockwise vortices that flow from the bottom to the top of the MP, driven by the recoil pressure, are indicated by the yellow arrow. The counterclockwise vortices that flow from the central region of the MP to the periphery, driven by Marangoni convection, are indicated by the green arrow. The key flow direction, which is directly related to the difference in bubble formation phenomena, is indicated by the black arrow. (Laser power= 312W , $t_{\text{off}}=t_3\mu\text{s}$).

3.3. Influence of laser exposure time on depth ratio and bubble formation

As shown in Fig. 11, the bubble formation phenomenon for laser exposure times of $90\mu\text{s}$ (*C1*), $105\mu\text{s}$ (*C2*), $125\mu\text{s}$ (*C3*), and $130\mu\text{s}$ (*C4*) under constant laser power (312W) and beam diameter ($140\mu\text{m}$) is investigated. At $90\mu\text{s}$, with a depth ratio of 0.732, laser removal causes the MP free surface to contract upward by surface tension and Marangoni forces, as shown in Fig. 11a. When the depth ratio increases to 0.837 at $105\mu\text{s}$, the removal of laser results in the formation of a small bubble within the MP at $108\mu\text{s}$ (Fig. 11b). If the laser is removed at $125\mu\text{s}$ (depth ratio=0.769 after keyhole fluctuations), the MP can contract upward smoothly without any bubbles (Fig. 11c). However, if the laser is off $5\mu\text{s}$ (depth ratio=0.813), a bubble is entrapped in the MP at $133\mu\text{s}$ (Fig. 11d) that is larger than that observed in Fig. 11b. It can be seen that in the keyhole mode, bubble formation exhibits stochastic behavior dependent on laser exposure time but predominantly occurs at extremely high depth ratios (cases *C2*, *C3*). This suggests a potential correlation between the melt pool depth ratio and the bubble formation mechanism when varying the laser exposure time.

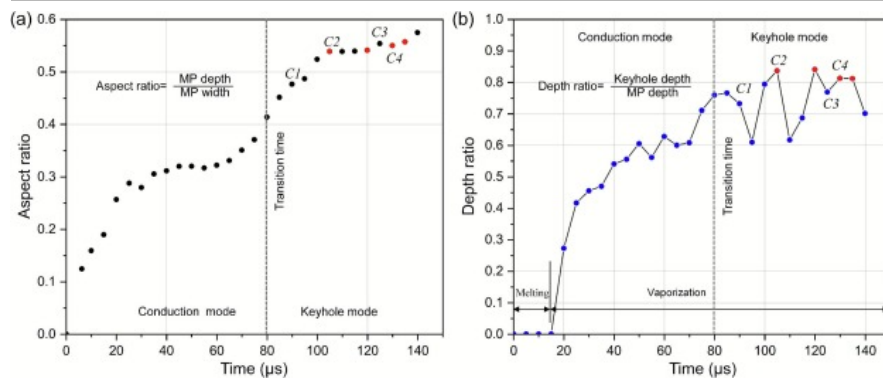


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Fig. 11. The keyhole morphology and liquid fraction distribution at one half of the simulation domain. The molten region (red, liquid fraction=1.0), the completely unmelted zone (blue, liquid fraction=0), and the mushy zone (liquid fraction between 0 and 1) are shown. The evolution of the keyhole profile and bubble formation phenomena within the molten pool is observed. (Laser power=312W, laser beam diameter=140μm). (a) C1: $t_{\text{off}}=90\mu\text{s}$, (b) C2: $t_{\text{off}}=105\mu\text{s}$, (c) C3: $t_{\text{off}}=125\mu\text{s}$, (d) C4: $t_{\text{off}}=130\mu\text{s}$.

As can be seen from Fig. 12a, the aspect ratio (MP depth/MP width) of the MP generally exhibits an increasing trend over time. Before the transition time, in the conduction mode, the aspect ratio increases at a relatively high rate. After the transition time, in the keyhole mode, the rate of increase in the aspect slowed when it reached a value of about 0.5. Furthermore, the laser is turned off at each point in the plot, corresponding to cases (including C1-C4) with different laser exposure times. The points are color-coded according to whether bubbles are observed subsequently: red markers indicate cases where bubble formation occurs, while black markers denote cases with no bubbles. As shown in Fig. 12a, bubbles occur randomly with laser exposure time and are independent of the aspect ratio. Fig. 12b shows the temporal evolution of the depth ratio, defined as the ratio of the keyhole depth to the melt pool depth. During the melting stage, the depth ratio is zero. In the vaporization stage, the depth ratio gradually increases in the conduction mode with minor fluctuations, while it exhibits significant volatility in the keyhole mode. Similarly, the occurrence of bubbles after laser-off at the corresponding moments is distinguished by color-coded dots. As illustrated in Fig. 12b, it is evident that bubble trapping is more probable at a higher depth ratio (cases C2, C4), represented by red dots, whereas no bubbles are trapped at a lower depth ratio (cases C1, C3), represented by blue dots. In other words, for different heating times, whether a bubble becomes trapped depends on whether the depth ratio is sufficiently high, revealing an inherent correlation between the depth ratio and the propensity for bubble formation. Essentially, during periods of intense keyhole depth fluctuation, the melt pool depth increases relatively slowly, leading to significant variations in the depth ratio. Simulation results demonstrate that bubbles are more likely to form under higher depth ratio conditions. It is precisely the pronounced fluctuation in keyhole depth that governs the dynamic variation of the depth ratio, thereby accounting for the absence of a clear correlation between bubble formation and heating time. Therefore, by incorporating the depth ratio as a key parameter, the analysis not only effectively captures the influence of keyhole depth fluctuations but also provides a rational explanation for the observed lack of a direct relationship between bubble formation and heating duration.



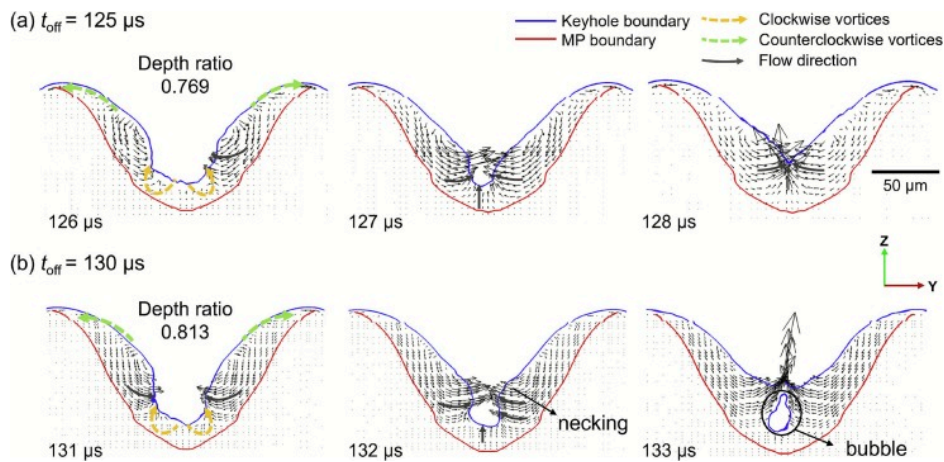
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Fig. 12. Temporal variation of MP morphology parameters. (Laser power=312W, $t_{\text{off}}=140\mu\text{s}$). (a) Temporal variation of aspect ratio (MP depth/MP width). (b) Temporal variation of depth ratio (Keyhole depth/MP depth). Bubble-trapping events for cases where the laser is shut off at that instant are distinguished by color: the red dot indicates the moments that, if the laser is off, bubble entrapment would occur, and the blue/black dot indicates that no bubble would form.

Fig. 13 shows the velocity field map at the melt pool cross-section $x=250\mu\text{m}$ with MP boundary and keyhole boundary for cases C3 ($t_{\text{off}}=125\mu\text{s}$) and C4 ($t_{\text{off}}=125\mu\text{s}$), to further investigate the mechanism linking depth ratio and keyhole-induced bubble formation phenomena. As shown in Fig. 13a, rapid velocity at the melt pool free surface decay (at $126\mu\text{s}$) and subsequent reversal upward (at $127\mu\text{s}$), while melt pool contraction leads to keyhole narrowing. These can also be observed for case C4 with a laser exposure time of $130\mu\text{s}$ (Fig. 13b). The critical distinction leading to different phenomena lies in the fact that the MP with a smaller depth ratio requires a shorter path for gas ejection, which is one reason why a larger depth ratio MP is more likely to trap bubbles. Moreover, it can be observed that there is a difference in the curvature of the keyhole sidewalls (Fig. 13, Fig. 13). A larger depth ratio tends to result in more vertically oriented sidewalls, which undergo horizontal contraction, as evident at $132\mu\text{s}$ in Fig. 13b, leading to pronounced necking. In summary, the depth ratio affects the different phenomena of bubble capture mainly through two factors: first, the length of the gas escape path at the bottom of the keyhole, and second, the direction of contraction of the keyhole sidewalls. Under the combined effect of these factors, a molten pool with a higher depth ratio is more prone to necking and subsequent bubble formation. Current findings confirm that in the keyhole mode characterized by minimal aspect ratio variation but significant depth ratio fluctuation, the magnitude of the depth ratio strongly correlates with bubble formation. Given the need to verify the broader applicability of this correlation, detailed validation under diverse processing conditions represents a critical next step and a major direction for future research. While the present work focuses on stationary laser, it is natural to ask whether the proposed correlations (e.g., depth ratio) are also found under moving laser. The thermal conduction and keyhole evolution mechanisms revealed herein exhibit general applicability, allowing for effective correlation with scanning laser processes. The scan speed serves as a critical parameter linking static and dynamic conditions: a lower scan speed corresponds to a longer local effective heating time, which in turn influences keyhole stability. This study demonstrates that a larger depth ratio is more likely to induce bubble formation. Based on this mechanism, the findings may offer new insights for optimizing scanning laser processes—for example, by enabling real-time adjustment of the scan speed through flexible monitoring of the melt pool morphology, thereby

controlling melt pool behavior and suppressing bubble-related defects. Such investigation will be the subject of a forthcoming work.



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Fig. 13. Velocity field in the cross-section $x=250\mu\text{m}$ for cases C3, C4 (Laser power=312W, laser beam diameter=140 μm). (a) C3: $t_{\text{off}}=125\mu\text{s}$, (b) C4: $t_{\text{off}}=130\mu\text{s}$. The red line (liquid fraction=0.5) indicates the liquid–solid interface, i.e., the MP boundary, while the blue line ($\alpha_1=0.5$) marks the keyhole boundary. The clockwise vortices that flow from the bottom to the top of the MP, driven by the recoil pressure, are indicated by the yellow arrow. The counterclockwise vortices that flow from the central region of the MP to the periphery, driven by Marangoni convection, are indicated by the green arrow. The key flow direction, which is directly related to the difference in bubble formation phenomena, is indicated by the black arrow.

4. Conclusion

In this work, the keyhole closure dynamics and bubble formation dynamics under stationary laser irradiation are investigated by a high-fidelity multi-physics CFD model. The following conclusions can be drawn from this study:

- (1) During the stationary laser irradiation, a transition from conduction mode (a quasi-semicircular profile) to keyhole mode (a spike profile) may occur depending on laser intensity. The bubble formation phenomenon is absent in conduction mode but may occur in keyhole mode under certain conditions. With a low surface tension, the keyhole becomes unstable and prone to collapse due to the intense melt flow, which enables bubble formation. However, when surface tension is sufficiently high, the melting remains in conduction mode via delaying the transition, thereby preventing bubble formation.
- (2) The transition, governed by the balance between surface tension and recoil pressure, marks the onset of keyhole instability. Simulation results reveal that increasing surface tension or lowering recoil pressure each extends the transition time, thereby suppressing bubble formation.
- (3) For different laser beam diameters involving keyhole mode, it was found that bubble formation is more likely to happen under smaller-diameter beams via a higher aspect ratio. A high aspect ratio, with narrow and deep MP, promotes necking by facilitating the collision and impingement of clockwise and counterclockwise vortices. However, with a lower aspect

ratio, vortices will not collide due to the wide, shallow MP, avoiding the formation of bubbles.

- (4) As laser heating time increases under fixed laser beam diameter and power in keyhole mode, the keyhole starts to experience intense fluctuations. During this keyhole mode, the aspect ratio gradually stabilizes, losing its correlation with bubble formation. The influencing parameter shifts to the depth ratio. High depth ratios, characterized by steep keyhole walls and extended collapse lengths, facilitate bubble formation. It is worth emphasizing that the newly proposed parameter, depth ratio, exhibits significant relevance during stages where the aspect ratio stabilizes and the keyhole undergoes intense fluctuations, thereby compensating for the limitations of aspect ratio characterization.
- (5) The research introduces a process parameter optimization strategy utilizing real-time aspect ratio and depth ratio monitoring, thereby presenting a potential pathway for actively suppressing bubbles in engineering applications. Real-time tracking of melt pool and keyhole morphological features enables early prediction of bubble generation trends, thereby allowing parameter optimization to minimize pore defects and enhance the densification quality of fabricated components. In terms of materials, the keyhole dynamics and melt pool instability mechanisms we identified are expected to be highly relevant for other materials with similar properties, such as 316L stainless steel and IN718 nickel-based superalloy, which also exhibit low thermal conductivity and a negative surface tension coefficient.

CRedit authorship contribution statement

Shiyu Liu: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Siyu Li:** Software, Methodology, Investigation. **Chenyu Lu:** Software, Methodology, Investigation. **Dongmin Wang:** Writing – review & editing, Supervision. **Xiaojun Quan:** Supervision, Resources.

Declaration of competing interest

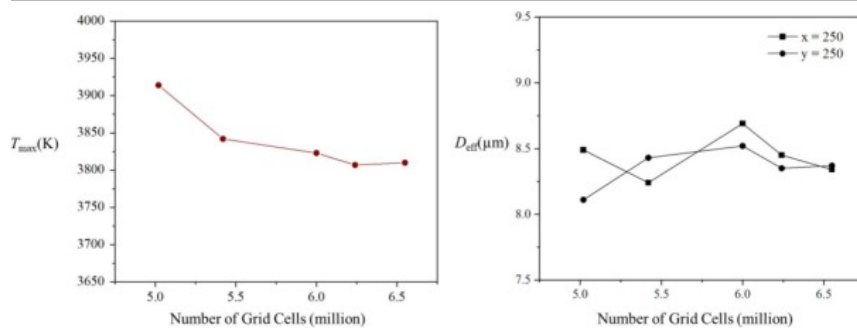
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

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Appendix A. Verification of grid independence

To demonstrate that the simulation results are independent of the computational cell size, we have performed a grid independence verification, as shown in [Fig. A1](#).



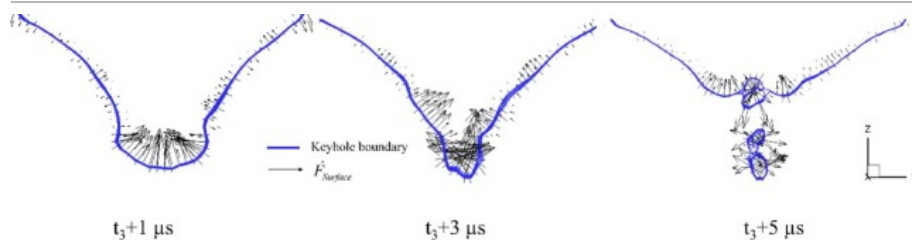
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Fig. A1. Verification of grid independence. (a) Variation in the maximum temperature across the entire computational domain at time t_1 with different grid systems; (b) evaluation of the equivalent bubble diameter at the cross-section ($x=250\mu\text{m}$, $y=250\mu\text{m}$) under various grid refinements.

Appendix B. The surface tension and recoil pressure variation in the necking region

Fig. A2 shows the change of surface tension variation in the necking region corresponding to Fig. 10. As shown, the surface tension only exists in the region of the gas–liquid interface, pointing in the direction of the curvature center of the curved interface, and its magnitude is closely related to the value of the curvature.



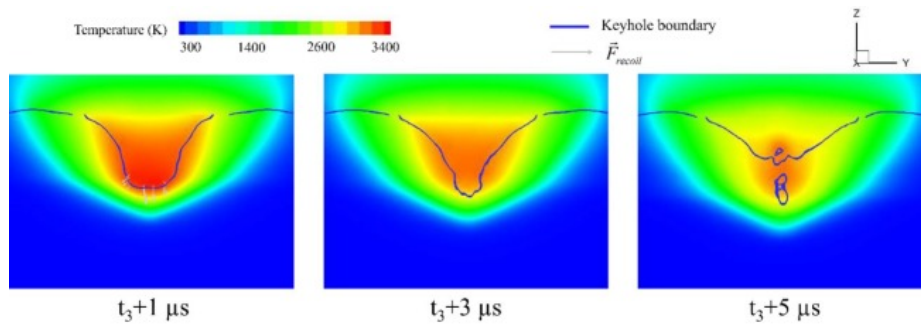
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Fig. A2. The surface tension variation in the necking region corresponding to Fig. 10. Surface tension (black arrows) and keyhole boundary, i.e., the gas liquid interface (blue line).

At $t_3\mu\text{s}$, the heat source is deactivated, and the temperature begins to decrease rapidly. However, the central region of the molten pool remains above the evaporation temperature (3313K) at $t_3+1\mu\text{s}$, so the recoil pressure is still present in this area, as shown in Fig. A3. By $t_3+3\mu\text{s}$, the surface temperature of the molten pool continues to drop, resulting in the temperature falling below the evaporation temperature at the location where necking occurs. As a consequence, the recoil pressure diminishes to zero. With

further cooling beyond $t_3+3\mu\text{s}$, the region is no longer subjected to recoil pressure.



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Fig. A3. The recoil pressure variation with temperature field corresponding to Fig. 10. Recoil pressure (gray arrows) and keyhole boundary, i.e., the gas liquid interface (blue line).

Data availability

Data will be made available on request.

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